

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of B Sc. Biological Sciences Examination November 2018

BS 102 Concepts of Physical Chemistry

Date: 19/11/2018 Day: Monday Time: 10.00 am To 10.20 am Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. [10]

- 1 Relation between enthalpy (H), pressure (P), volume (V), and internal energy (E) is given by:
(a) $\Delta G = \Delta H + T\Delta S$ (b) $\Delta G = \Delta H - T\Delta S$
(c) $\Delta G = \Delta H + \Delta S$ (d) $\Delta G = T\Delta H + \Delta S$
- 2 The sum of power of concentration term in rate law equation is called
(a) Rate of reaction (b) Rate constant
(c) Equilibrium (d) Order of reaction
- 3 The optical density of color solution increases with _____ in concentration
(a) increases (b) decreases
(c) does not change (d) may increase or decrease
- 4 The rate constant of a reaction is found to be $0.01 \text{ Mol}^1 \text{ li}^{-1} \text{ s}^{-1}$, the order of the reaction is:
(a) 1st (b) 2nd
(c) Pseudo 1st (d) zero
- 5 The process in which $\Delta P = 0$ is called
(a) Isobaric (b) Isochoric
(c) Isothermal (d) Adiabatic
- 6 A straight line parallel to Y axis has slope :
(a) zero (b) positive
(c) Infinity (d) negative

- 7 The correct relation for half-life time for first order chemical reaction is:
- (a) $K = 0.693 / t_{1/2}$ (b) $K = t_{1/2} / 0.693$
(c) $K = 0.639 \times t_{1/2}$ (d) $[C_o] = K \times t_{1/2}$
- 8 The surface tension of the water will _____ when surfactant added
- (a) increase (b) decrease
(c) remain unchanged (d) become zero
- 9 For reaction to non-spontaneous, which relation is correct?
- (a) $\Delta G > 0$ (b) $\Delta G < 0$
(c) $\Delta G = 0$ (d) $\Delta G = \text{infinity}$
- 10 Find the **slope** (m) and **intercept** (C) of the following equation $4y + 8x - 12 = 0$
- (a) $m = 6, c = 3$ (b) $m = -2, c = 3$
(c) $m = 2, c = -6$ (d) $m = 2, c = 3$

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I Semester of B Sc. Biological Sciences Examination November 2018

BS 102 Concepts of Physical Chemistry

Date: 19/11/2018 Day: Monday Time: 10.20 am To 12.00 noon Maximum Marks: 25

Instructions:

1. **Section I and II** must be attempted in **SINGLE ANSWER SHEET**.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q - II Answer the following questions as directed. (Any Five) [10]

- 1 State and explain Charles's law.
- 2 20 g MgSO_4 (Mol. Weight = 120.3 g/mol.) is dissolved into 250 ml distilled water. Calculate the concentration of the solution in molarity.
- 3 For the reaction $\frac{1}{2}\text{N}_2 + \frac{3}{2}\text{H}_2 \rightarrow \text{NH}_3$, write the rate of the reaction with respect to H_2 and NH_3 .
- 4 Draw straight lines with (i) slope positive, (ii) slope Negative.
- 5 What is system? Write Different types of system.
- 6 Mathematical formula of surface tension. CGS and SI unit of surface tension.
- 7 Strong and weak electrolytes: example and definition

SECTION – II

Q-III Answer the following questions as directed. (Any Three) [15]

- 1 Show that, for one mole of ideal gas, $C_P - C_V = R$ where R is universal gas constant.
- 2 Define the following concentration: % (w/v), % (w/w), molarity, molality, mole fraction, normality
- 3 Explain in detail the kinetic theory of gas.
- 4 Derive the Arrhenius relation for determination of activation energy.
- 5 What is surface tension discuss with proper diagram?